AI Adoption in Ride – Hailing Industry: EV Fleet Operation Optimization

Prof. Ahmed ElSeddawy
Vice Dean of Educational Affairs- Smart Village
Ahmed.Bahgat@aast.edu

Eng. Mohamed Tarek Ragab
Chief Technical Officer – Udexon
Mohamed.Tarek.Ragab@gmail.com

1. Introduction

Shifting towards electric vehicles (EVs) is one of the challenges that is facing the automotive sector, which is mainly driven by the increasing the awareness for the environmental risk due to using traditional transportation. on the other hand, transition towards the evolution of electric vehicles requires creative solutions to ensure the maximum utilization and efficiency of electric vehicles ("IOT-ENABLED FLEET MANAGEMENT AND ITS INFLUENCE ON COST REDUCTION AND PROFITABILITY," 2026).

The integration of Artificial Intelligence and the Internet of Things form Artificial intelligence of things (AIoT) drastically transforms fleet operation management by shifting operations from reactive, manual processes to dynamic, data-driven systems. AIoT impacts efficiency across several core areas of fleet management.

The Artificial Intelligence of Things (AIoT) is the migration of Artificial Intelligence (AI) models with internet of thins and its sensors network ("IOT-ENABLED FLEET MANAGEMENT AND ITS INFLUENCE ON COST REDUCTION AND PROFITABILITY," 2026). In electric vehicle (EV) fleet management, an AIoT framework involves installing embedded IoT sensors in vehicles to continuously capture real-time telemetry data such as battery voltage, thermal fluctuations, and spatial coordinates then processing and analyzing this continuous data stream through advanced machine learning algorithms ("IOT-ENABLED FLEET MANAGEMENT AND ITS INFLUENCE ON COST REDUCTION AND PROFITABILITY," 2026). This integration creates a data-driven and dynamic, autonomous ecosystem capable of translating raw telemetry vehicle data into intelligent system by enabling predictive maintenance diagnostics, dynamic route optimization, and enhanced energy utilization across the entire EV fleet (Bahheti et al., 2024).

1.1. Motivation

This paper examines the implementation of artificial intelligence (AI) in the optimization of electric vehicle (EV) fleet performance, with clarifying the impact and consequences of this adoption on Operational Expenditure (OPEX) and the broader optimization of Total Cost of Ownership (TCO).
Furthermore, the paper is focusing on the integration of AI-driven dynamic routing and dispatching systems for EVs, alongside a proposed framework for the digital transformation and the systematic migration of AI within business operations.

1.2. Methodology of Decision Making

The process of decision-making to adopt AI and related computerized support systems in an organization involves a structured sequence of stages, starting from the identification of a business problem to the evaluation of AI feasibility and finally managing user adoption.

The General Decision-Making Framework introduced in (Averweg, n.d.) is presenting a process typically follows a sequential decision-making model. The proposed framework involves four main stages, as shown in Figure 1:

- **Intelligence:** it is the step of discovering, identifying, and understanding the problems occurring within the organization and why they exist.
  through our study we will use Data analytics with large amounts of datasets to monitor business operations of EV Fleet, explore hidden patterns and define why a problem exists falls into this stage and proof the problem existence using evidence from DA.
- **Design:** Identifying, inventing, and exploring different potential solutions to the available problems.
  A comprehensive and literature review across academic and industry reports to achieve existing approaches and identify practices for the adoption of artificial intelligence in electric vehicle fleet operations and management.
- **Choice:** In this step, we are selecting the best fit solution for our problem across the available alternatives. Where the decision was guided by recommending the solution that has the greatest potential impact aligned with the organization's strategic objectives and operational capabilities.
- **Implementation:** Finally implementing and deploying the target solution with the support of the necessary digital transformation to enable effective AI integration with the organization. The implementation includes integration with existing Enterprise Resource Planning (ERP) organization's platform.

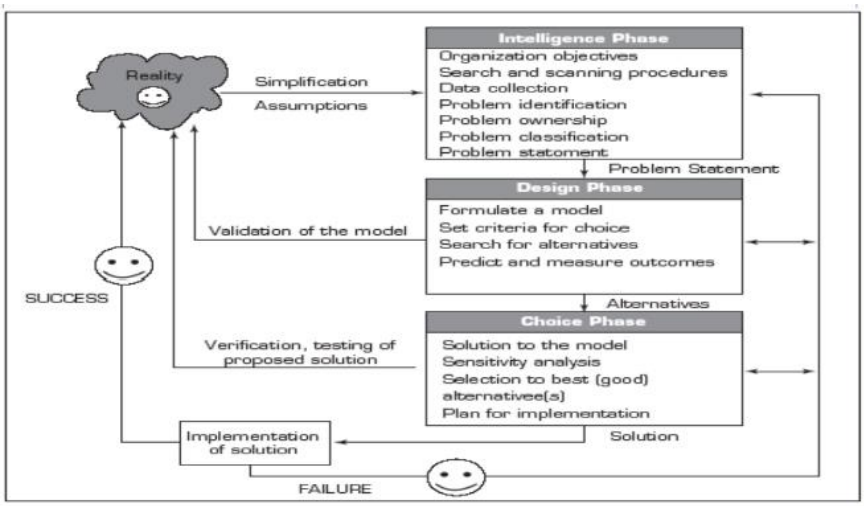

Figure 1 The Decision Making/Modeling Process

2. Business Analysis and Intelligence

2.1. Industry Overview & Business Context

The ride-hailing industry, a core component of the broader Mobility as a Service (MaaS) sector, leverages digital platforms to automate the matching of passengers with available drivers or vehicles as presented in the industry report reference (Shi et al., 2020).

The global ride-hailing industry is a rapidly growing component of shared mobility, expected to grow from an estimated USD 158.3 billion in 2023 to over USD 480 billion by 2032 (Ltd, n.d.).

Figure 2 Ride-Hailing Market Analysis *(Ltd, n.d.)*

According to (*Secrets Of The Ride Hailing Business Model | Appscrip Blog*, 2024) Ride-hailing platforms operate using several distinct business models, often mixing them to increase market penetration, Table 1 shows a comparison between different business models:

Business Model	Description	Key Characteristics
Pure Self-Operated Platform (PSP)	A centralized operational model in which the platform directly manages the interaction between passengers and drivers.	Drivers typically use their own vehicles; the platform controls dispatching, pricing, and payment processing.
Pure Aggregation Platform (PAP)	An intermediary marketplace model that does not directly manage drivers or vehicle fleets.	Aggregates local ride-hailing providers; revenue is generated through commission-based or franchise fee structures.
Self-Operated + Aggregation Platform (S+A-P)	A hybrid model combining direct fleet ownership with the aggregation of third-party service providers.	Enhances coverage and operational capacity through both in-house and outsourced fleets.
Super App Model	An integrated digital platform that extends beyond ride-hailing services.	Common in Asian markets; includes food delivery, logistics, and digital financial services.

Table 1 Ride-Hailing Business Models

The transportation industry is currently experiencing an exponential growth navigating a transition and shifting towards electric vehicle fleets (EVs), facilitated by industry leaders such as Uber, Lyft and DiDi (Shi et al., 2020).

This shift is fundamentally driven by the dual objectives of mitigating global carbon emissions as part of Sustainable Development Goals of 2030 as shown in the following figure 3 (*THE 17 GOALS | Sustainable Development*, n.d.). Beside to optimizing the Total Cost of Ownership (TCO), as EVs offer significantly lower long-term operational and maintenance expenses compared to traditional widely used internal combustion engine (ICE) vehicles (Shi et al., 2020).

Figure 3 UN SDG Goals

Shifting from internal combustion engine (ICE) vehicles to EVs isn't just a one for one swap, it faces new logistical considerations and challenges around charging, vehicle deployment and energy management. Focusing on the EV fleet management is important because it helps fleet operators navigate these changes and gain the benefits of electrification.
if transformation and management is done right, it goes beyond basic replacement and enables significant cost savings. Intelligently replanned systems have shown 8%–13% lower total cost of ownership (TCO) at the fleet level compared to tradition Internal combustion engine operations, in contrast to a simple 1:1 swap by just replacing ICE vehicle with Ev yielded just 3% savings, heavily reliant on CAPEX subsidies (*EV Fleet Management Made Scalable and Efficient – gridX*, n.d.).

Target Organization:
Going through the study we will focus on an organization in the ride-hailing industry with following specifications:

a. Organization Business Model is Self-Operated + Aggregation Platform (SAP).
b. The organization owns and operates a **fleet of 57** Electric vehicles from different OEM and Manufacturing years.

2.2. Business Problem Identification

With close focus on the underlined ride-hailing organization, we have identified the following pain points and concerns to be proved through descriptive analysis of the fleet operational dataset.

The organization is suffering of poorly planning of dispatching and routes of the vehicle and discard the impact of traffic, weather conditions and cargo loads. Which led to non-profitable energy consumption due to the long empty distances to client pickup points "Dead Range" and not considering the route dynamics and updates, impacting the Operational Expenses and the Total Cost of Ownership (TCO).

Another side effect for the poor dispatching is the high cancellation rate of the trips with long waiting times of the driver who is covering a long pickup distance, leading to high customer churn and decline in the profitability of the organization.

Problem Statement: Traditional fleet management is reactive. Decisions made lead to high operational costs and poor fleet utilization and management.

Business Challenges and Pain Points: High operational costs (OPEX), "Range Anxiety", Dead Range Optimization, Dynamic Dispatching and VRP.

VRP stands for the Vehicle Routing Problem, which is an operational optimization framework utilized to determine the most efficient routes and dispatch schedules for a fleet of vehicles (Aldossary, 2024).

2.3. Traditional Solutions Vs Intelligent Decision Making

Several challenges that made traditional or systems (like model-based optimization or simple heuristics) difficult to use for solving the electric vehicle (EV) fleet dispatch problem:

a. The unknown customer trip requests and additional complexity of recharge demands make it difficult to develop a model-based optimization algorithm to solve this dynamic VRP (Shi et al., 2020).
b. The key difference between dispatching conventional gasoline vehicles and EVs lies in the relatively frequent recharging processes of EV batteries (Shi et al., 2020).

2.4. Business Operation and Financial GAP

One of the biggest operational gaps and challenges facing electric vehicle (EV) fleet operators today is figuring out how to respond, in real time, to the unpredictable nature of passenger demand while also navigating the logistical complexity of keeping vehicles charged — something traditional dispatch systems were simply never built to handle (Shi et al., 2020).

The financial toll of getting this wrong is very real. When routing decisions are poorly optimized, vehicles end up spending a significant portion of their time on the road with no passengers, either repositioning or hunting for an available charger. All those empty miles quietly drive up electricity bills and accelerate wear and tear on the vehicles themselves (Yi & Smart, 2021). According to industry reports , the "Dead Distance" losses reached to 25% of the total energy utilization done by the EV Fleet.

3. Data Analytics

3.1. Methodology

A quantitative analytical methodology was followed through this study, comprising the following sequential stages as show in Figure 4:

(a) Data Collection: Operational data related to the electric vehicle (EV) fleet were collected from the ride-hailing cloud platform responsible for fleet management operations. This dataset was merged by data obtained from IoT-based telematics devices installed in electric vehicles, which capture vehicle-level information as snapshots and transmit it to the cloud via wireless communication technologies like 4G.

(b) Data Cleaning and Modeling: Exploratory Data Analysis (EDA) techniques were applied to address data quality issues, cleaning raw data, including the treatment of missing values, removal of duplicate records, and correction of inconsistencies within the dataset.

(c) Visualization and Interpretation: Generate statistical visualizations, such as charts and histograms, to facilitate data interpretation, with particular attention given the organization problem identified from business analysis.

(d) Insight to Action: Analytical findings like integrating algorithms and solutions, and provide insightable actions to support evidence-based decision-making and operational improvements.

(e) Iterative Refinement: For confirming the model continuous trusted output, The data, analytical models, and key performance indicators (KPIs) were continuously refined to enhance model accuracy and ensure sustained performance improvement.

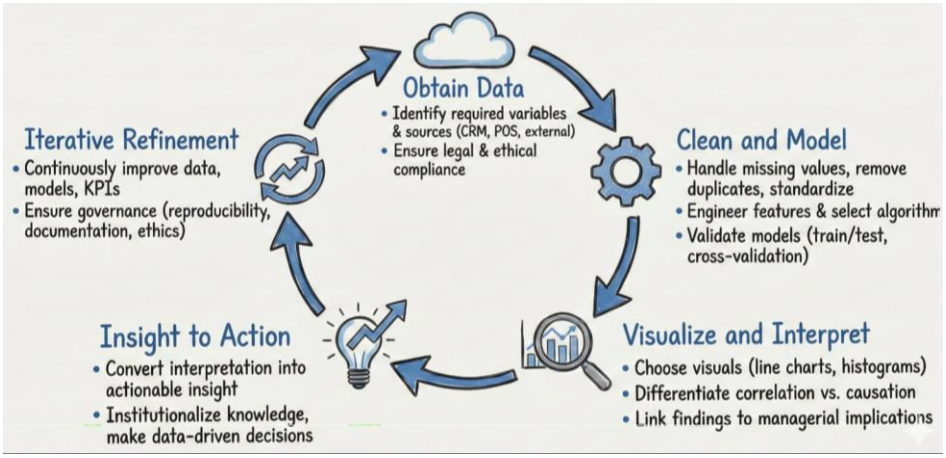

Figure 4 Data Analytics Framework

3.2. Dataset Overview

The Electric Mobility Ride Dataset was collected from Kaggle as combination of (*Electric Mobility Dataset*, n.d.) and (*Electric Vehicle Analytics Dataset.*, n.d.), where (*Electric Mobility Dataset*, n.d.) offers a rich and detailed snapshot of electric mobility ride transactions, capturing key aspects of user behavior, driver performance, fare details, and operational metrics contains 2499 rides with total activity

duration of 18 Months from Jan 2023 to July 2024 including multiple rides per day.
While (*Electric Vehicle Analytics Dataset.*, n.d.) is a synthetically dataset provides a comprehensive view of electric vehicle (EV) performance, charging behavior, battery health, and cost analytics across different manufacturers, models, regions, and usage types.

The Dataset is classified as **structured Realtime** data collected from ride-hailing cloud platform in collaboration with IoT-based telematics devices installed in 57 Electric Vehicles for

3.3. Data Analytics Interpretations and Insights

Customer Waiting Time Interpretation:
Figure 5 provides a detailed look at the fleet's pickup response time for customer requests as with a heavily right-skewed distribution, with long tail extending up to 60 minutes.

This indicates that part of the passengers are experiencing long waiting times up to 60 minutes.
Based on the analysis of the ride data, **17.65%** of all rider (441 out of 2,499) had a waiting time greater than 15 minutes.

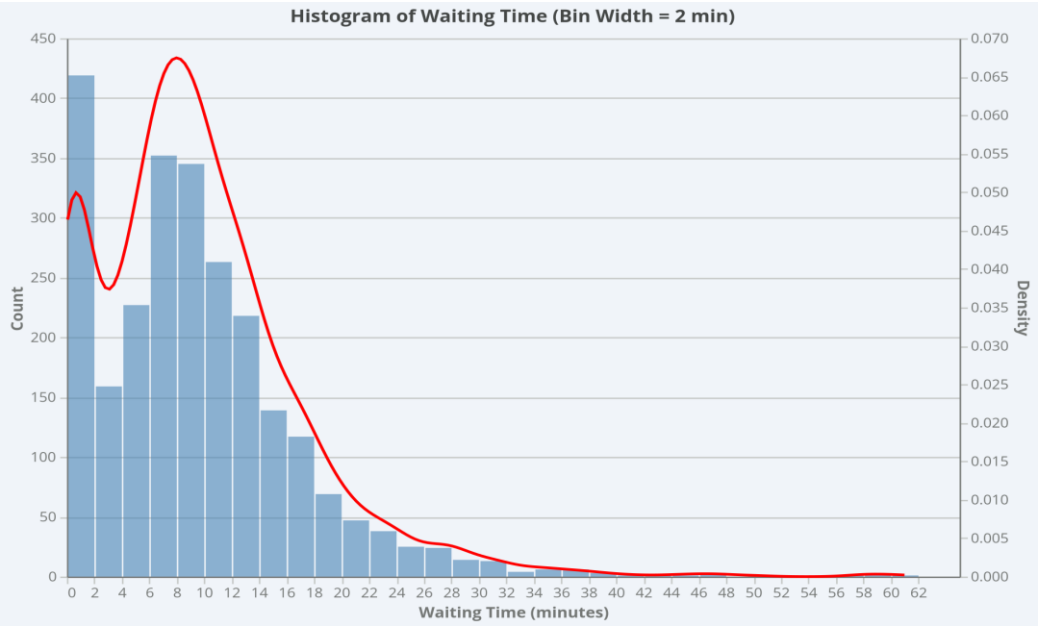

Figure 5 Waiting Time Distribution

Cancelled Trips Root Cause Interpretation:
The relationship between the distribution of cancelled trips and the maximum ride accept distance per hour is clarified in Figure 6. This chart highlights the extreme cases where driver proximity and distance travelled for pickup may have directly influenced the decision to cancel a trip.
During the evening peak (from 17:00 to 19:00), some users are being matched with drivers from very far away (Dispatching Problem) , significantly increasing the likelihood of cancellation.

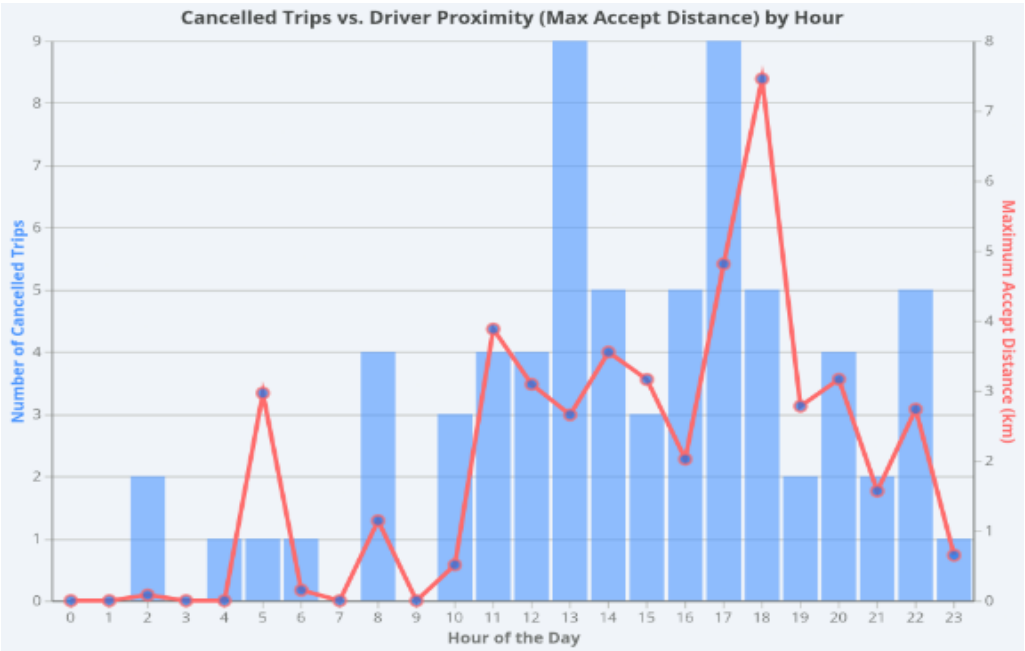

Figure 6 Cancelled Trip Vs Dead Distance

Dead Energy Interpretation:

Across all vehicles, by linking the vehicles information randomly (57 EVs) with the data rides, a calculation of Dead Energy accounts for approximately 25% of the total energy consumed by the fleet for dead distance as shown in Pie Chart in Figure 7.

This means nearly a quarter of the fleet's energy is spent before a passenger even enters the vehicle. This significant high ratio indicates they are being dispatched to passengers who are far away or that they operate in areas with lower ride density.

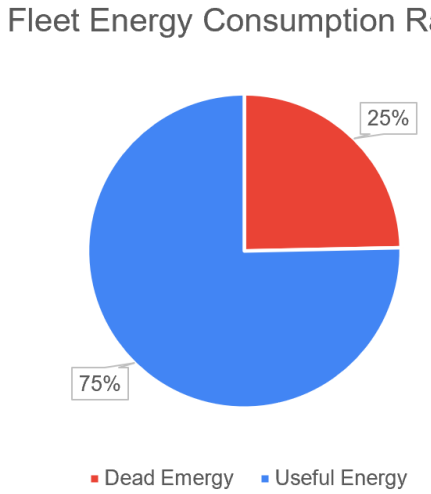

Figure 4 Wasted Energy Ratio

3.4. Insights and Conclusions

Based on the preceding analysis, several critical insights have been identified. Approximately 20% of all completed rides facing passenger waiting times ranging between 15 and 60 minutes, which is significantly higher the ride-hailing industry benchmark of a maximum 15-minute pickup time and average 7 minutes.

Furthermore, a strong correlation between trip cancellations and increased ride acceptance distance indicating the limitations of the current existing dispatching methodology, particularly during peak demand periods.

The analysis also reveals that nearly 25% of total electric energy consumption is directed towards non revenue generating operations, indicating substantial inefficiencies that affects negatively both energy sustainability and carbon emission reduction objectives.

In conclusion, optimizing EV fleet operations through advanced dispatching policies and intelligent dynamic routing algorithms is essential to reducing customer **waiting times**, **mitigating range inefficiencies**, and **minimizing wasted energy**.

By enabling more accurate and precise vehicles to passengers matching through allocation to the nearest available driver, fleet efficiency can be significantly enhanced.
The subsequent sections will therefore examine the role of AI in addressing these identified operational challenges and minimizing the associated business pain points.

4. AI Adoption in Dynamic Dispatching & VRP

Our problem belongs to the dynamic Vehicle Routing Problem (VRPs) as the customer trip requests are not known in advance but received during the operating process (Shi et al., 2020). Solving Dispatching and allocation of correct EV to correct customer problems shall be discussed using **Reinforcement Machine Learning Model**.

4.1. Literature Review For AI in EV Operations

In this section we will show different areas where AI can be adopted and provide impact on the Operation of Electric Vehicles Fleet.

As presented in (Patel & Paldhikar, 2025), Predictive Maintenance and Minimized Downtime systems are integrating real-time data from IoT sensors and get the advantage of advanced machine learning models such as Random Forest, XGBoost, and Long Short-Term Memory (LSTM) networks to detect and anticipate system failures before they occur.

Enhanced Driver Monitoring and Safety presented in ("IOT-ENABLED FLEET MANAGEMENT AND ITS INFLUENCE ON COST REDUCTION AND PROFITABILITY," 2026) Through AI & IoT telematics as fleet managers can continuously monitor driver behavior, detecting actions such as over speeding, harsh braking and rapid acceleration. AI and machine learning modules evaluate this behavioral data to generate predictive safety alerts and provide targeted driver coaching.

In Energy Management and Cost Reduction For electric vehicle (EV) fleets, AI & IoT acts as a critical tool for controlling energy consumption. In EV fleets, AIoT monitors battery health, energy usage, and route profiles to intelligently recommend the most cost-effective charging schedules, such as charging during off-peak electricity hours ("IoT Solutions for Real-Time Monitoring and Management of Electric Vehicle Fleets," n.d.).

The writer in ("IoT Solutions for Real-Time Monitoring and Management of Electric Vehicle Fleets," n.d.) mentioned the impact of Artificial on Dynamic Routing and Dispatch Optimization for the enhancement of dispatching by continuously analyzing telematics data alongside external variables like traffic congestion and weather conditions.

4.2. AI for Dynamic Routing and Intelligent Dispatching

Reinforcement learning (RL) is explained in (Shi et al., 2020) as a computational approach to learning whereby an agent learns to make decisions by interacting with an environment.

Components of reinforcement learning:

- Agent and Environment: The vehicle (learner or decision-maker) is called the agent, which interacts with its environment over a sequence of discrete time steps.
- States, Actions, and Rewards: At each time step, the agent senses the environment's current state, takes a specific action, and subsequently receives a numerical reward based on that state-action pair.
- Goal of Maximization: The agent's objective is to discover an optimal policy (a mapping from states to actions) that maximizes the total discounted return (the sum of cumulative rewards over time).
- Value Functions: RL uses mathematical functions to evaluate the desirability of being in a particular state (state-value function) or taking a specific action from a state (action-value function).

Learning Types:

- On policy: The agent learns through the same policy it is currently using to make decisions.
- Off policy: The agent learns from data generated by a different "behavior" policy, allowing it to exploit historical data or data from other non-learning methods.

4.3. AI Proposed Solution

Shi et al. (Shi et al., 2020) proposes a reinforcement learning framework for EV fleet operation that integrates decentralized learning with centralized decision making as shown in Figure 8.
Individual EVs act as learning agents that interact with environment and share their experiences to address scalability challenges in multi agent environments.
Coordination and experience sharing between fleet vehicles is handled through a centralized mechanism that formulates dispatching as a linear assignment problem, enabling efficient solution by forming the problem within a Markov decision process (MDP) formulation.

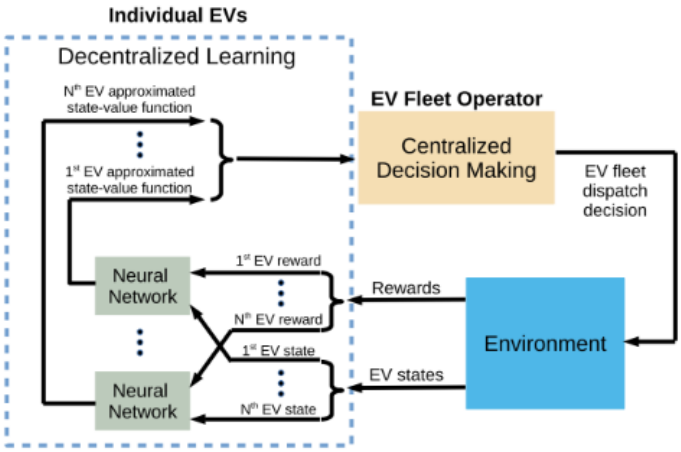

Figure 5 EV Centralized-Decentralized RL Model

In study (Shi et al., 2020) employs an off policy reinforcement learning algorithm in which the value function is approximated using a neural network. Reinforcement learning methods are distinguished by whether the behavior and target policies are identical, Where off policy approaches offering greater flexibility by learning from data generated under different policies. This enables more general learning and allows the effective use of historical data or data produced by non learning methods.

4.4. AI Adoption Conclusion

After running simulations, the results of from a comprehensive numerical study stated that the proposed reinforcement learning (RL) algorithm efficiently learns optimal EV fleet dispatching strategies and outperforms benchmark methods by reducing overall costs, including operational expenses and passenger waiting times (Shi et al., 2020).

The proposed RL-based EV fleet dispatch method achieves a **18.12%** and a **3.20%** cost saving on the testing samples compared with the normal dispatch algorithms (if they are used) as greedy algorithm and the optimization-based method (Shi et al., 2020).

Unlike static routing, AI-integrated systems can re-optimize routes as customer trip requests are revealed in real-time, making them well-suited for dynamic vehicle routing problems (VRPs) (Shi et al., 2020).

5. Digital Transformation and AI Implementation

The fleet management software integrates technologies such as cloud computing, data analytics, and connectivity, serving as the foundational layer for online services. On the other hand, fleet as a service (FaaS) operates as the consumer-facing layer, where end-users physically utilise fleets, accommodating both B2B and B2C models (*Electrification-of-Fleet-Operations.Pdf*, n.d.).

The Information system of the organization shall be updated to enable the migration and integration of a software platform for operating and managing the fleet.

According to KDD Pyramid in Figure 10, the following updates and adjustment should be deployed to enable the migration of AI to the current Organization's technology structure.

In The Infrastructure Technology layer is the physical that makes all the system works.
It requires to add GPU/TPU clusters like NVIDIA H100 for training large models and serving inference and operations at scale. For cloud and platform, The organization needs scalable cloud environment like AWS or Azure.

In the Data Layer, new components shall be added for managing data collection and interaction with the IoT telemetry devices and storing them as Warehouses, Big Data for enabling Machine learning and data mining. Enforcing Data governance and quality standards, investing and curation of datasets for AI training and processing.

The Information layer, real time processing, BI & Visualization Platforms and vector databases are needed.
A sample for dashboard can be show in the next (figure 9) for operating and managing Fleet of EVs.

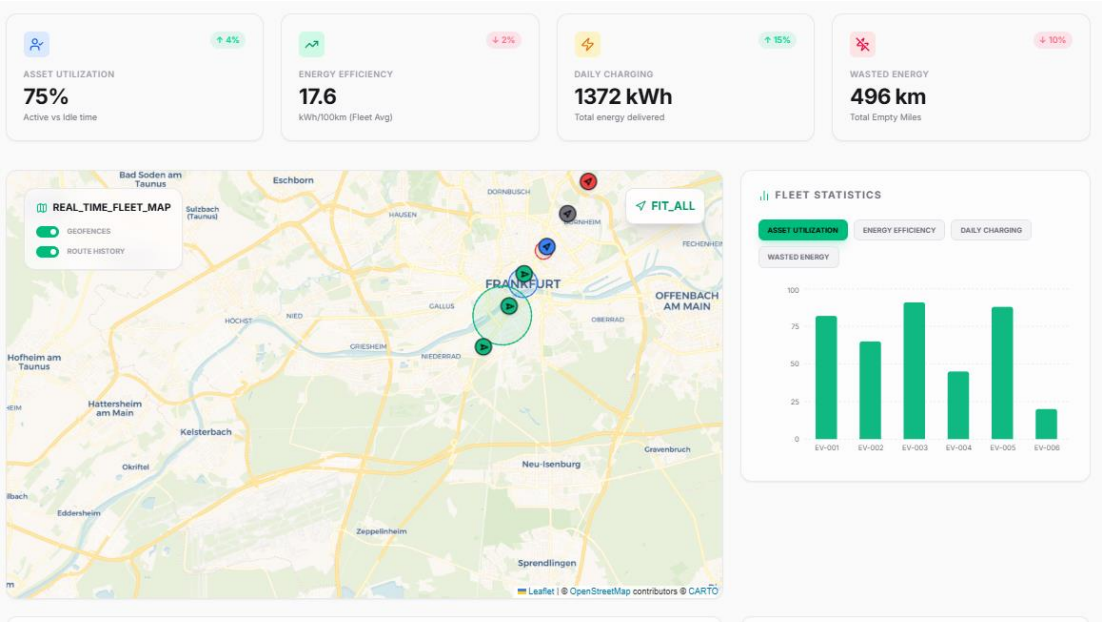

Figure 6 EV Fleet Operation Platform *(My Google AI Studio App, n.d.)*

The knowledge Layer where the ML models, LLMs and RAG system are running. In this layer the implementation of ML models and framework shall be achieved using PyTorch, TensorFlow, JAX.

Finally, the wisdom layer shall be upgraded to cover AI ethics & Governance policies, Strategic Decision Making.

Figure 7 KDD Technology Structure

6. ERP & RPA

The ERP of the fleet drivers shall be connected to AI models, to get insights about their performance, rating and attitude, that helps in the annual appraisals of the driver and supports in the stuffing decision making and releasing by HR Business Function, based on the performance prediction and business needs.

ERP modules Operation Depts, can get the benefit of AI integration in the prediction of the utilization of the fleet according to historical peak hours, and predict the failure of the vehicles and their forecasted capacity,

Robot Processing Automation shall automate the interaction between the AI and ERP system, where it acts as a critical bridge of connecting the rigid Enterprise Reousrce Planning system with Intelligent AI insights.

RPA can cross-reference ERP data with external databases, invoices and CRM records with daily reports. RPA can close the distance and GAP between the purchasing team, maintenance team and middle management where it follows a pre-set business rules.

7. Risks and Limitations of AI Adoption

The Adoption of AI in EV Fleet Operations is mainly impacting ISO/IEC 27001 Clause 4.1 and Clause 6.1. where Clause 4.1is requiring to understand the organization and its context either external issues like the EU AI Act and DORA , or the internal issues such as Shadow AI and resource constrains specific to fleet management. On the other hand, Clause 6.1 is focusing on actions to address risks and opportunities requiring a necessity for assessing new risks related to AI-specific threats like model poisoning, prompt injection, and adversarial attacks on fleet autonomous operations.

8. Strategic Insights & Recommendations

Today's market characteristics in the ride hailing industry attracting consumer attention are as follows: One of such strategies linked with market characteristics is the proliferation of EVs and AVs. Popular ride-hailing platforms are ramping up investments in EV portfolios to support ES goals and decrease organizational costs. Governments and market stakeholders are making this change possible by providing incentives such as subsidies and tax credits. Moreover, the number of fully self-driven cars in the ridesharing vehicles are also projected to empower the market by cutting dependence on the drivers, safety features and decrease in operational costs. While major participants have been testing experiments and case studies of self-driving automobiles in ride-booking services, the future landscape of ride-hailing is gradually autonomous and eco-friendly (*THE 17 GOALS | Sustainable Development*, n.d.).

By increasing the efficiency of EV operations, AI supports the reduction of greenhouse gas emissions and zero tailpipe pollution which has Environmental Impact and achieves the Sustainable Development Goals (SDG) of united nations.

9. Keywords & Definitions

- AIOT: Artificial integration of thins is, is the migration of IoT Embedded systems with AI models to collect, process ana analyze real-time data, which enables autonomous data driven decision making.
- SDG: a group of global objectives, settled by United Nations aimed at promoting environmental sustainability, including reducing carbon emissions through the adoption of electric mobility and intelligent transportation system.
- Reinforcement Machine Learning: A learning paradigm in which an agent interacts with its environment by taking actions and receiving rewards, with the objective of learning an optimal policy that maximizes cumulative long-term rewards.
- Random Forest: A machine learning ensemble method that constructs multiple decision trees and aggregates their outputs, commonly used in EV fleet operations for predictive maintenance and failure detection.
- XGBoost: A gradient-boosting machine learning algorithm optimized for high predictive accuracy and computational efficiency, applied in EV fleet analytics for system performance prediction
- LSTM Networks: A class of recurrent neural networks designed to learn long-term temporal dependencies, used for time-series analysis such as battery health monitoring and predictive maintenance in EV fleets
- Agent and Environment: The vehicle (learner or decision-maker) is called the agent, which interacts with its environment over a sequence of discrete time steps.
- States, Actions, and Rewards: At each time step, the agent senses the environment's current state, takes a specific action, and subsequently receives a numerical reward based on that state-action pair.

10. References

Aldossary, M. (2024). Enhancing Urban Electric Vehicle (EV) Fleet Management Efficiency in Smart Cities: A Predictive Hybrid Deep Learning Framework. *Smart Cities*, *7*(6), 3678–3704. https://doi.org/10.3390/smartcities7060142

Averweg, U. R. F. (n.d.). *Decision-making support systems: Theory & practice*.

Bahheti, M., Shankarnarayan, V. K., Durairajan, S., Chandak, S., Gurunathan, T., & Chandrasekar, P. (2024). AI-Powered Predictive Maintenance for Electric Vehicle Fleets. *2024 Asian Conference on Intelligent Technologies (ACOIT)*, 1–6. https://doi.org/10.1109/ACOIT62457.2024.10939072

Electric Mobility Dataset. (n.d.). Retrieved March 30, 2026, from https://www.kaggle.com/datasets/rakeshmeka/electric-mobility-dataset

Electric Vehicle Analytics Dataset. (n.d.). Retrieved March 30, 2026, from https://www.kaggle.com/datasets/khushikyad001/electric-vehicle-analytics-dataset

Electrification-of-fleet-operations.pdf. (n.d.). Retrieved March 22, 2026, from https://www.pwc.com/m1/en/publications/documents/2024/electrification-of-fleet-operations.pdf

EV fleet management made scalable and efficient – gridX. (n.d.). Retrieved March 25, 2026, from https://www.gridx.ai/knowledge/ev-fleet-management

IoT Solutions for Real-Time Monitoring and Management of Electric Vehicle Fleets. (n.d.). *International Journal of Scientific Research in Science and Technology*.

IOT-ENABLED FLEET MANAGEMENT AND ITS INFLUENCE ON COST REDUCTION AND PROFITABILITY. (2026). *International Journal of Progressive Research in Engineering Management and Science*. https://doi.org/10.58257/IJPREMS45155

Ltd, I. M. R. (IMR) P. (n.d.). *Ride-Hailing Market Overview and Growth Trends*. Introspective Market Research. Retrieved March 23, 2026, from https://introspectivemarketresearch.com/reports/ride-hailing-market/

My Google AI Studio App. (n.d.). Retrieved April 7, 2026, from https://udexon-fleetmanagmentdashbaord.onrender.com/

Patel, O., & Paldhikar, M. (2025). *AI-Driven Predictive Maintenance System for Smart Fleet Management*.

Secrets Of The Ride Hailing Business Model | Appscrip Blog. (2024, June 19). https://appscrip.com/blog/secrets-of-the-ride-hailing-business-model/

Shi, J., Gao, Y., Wang, W., Yu, N., & Ioannou, P. A. (2020). Operating Electric Vehicle Fleet for Ride-Hailing Services With Reinforcement Learning. *IEEE Transactions on Intelligent Transportation Systems*, *21*(11), 4822–4834. https://doi.org/10.1109/TITS.2019.2947408

THE 17 GOALS | Sustainable Development. (n.d.). Retrieved April 7, 2026, from https://sdgs.un.org/goals

Yi, Z., & Smart, J. (2021). A framework for integrated dispatching and charging management of an autonomous electric vehicle ride-hailing fleet. *Transportation Research Part D: Transport and Environment*, *95*, 102822. https://doi.org/10.1016/j.trd.2021.102822